Lecture 8: Sections 2.3, 2.4, and 2.5

Differentiability and Derivatives

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Differentiability: Definition

Intuitive Definition

A function is **differentiable** at a point if it has a well-defined, non-vertical tangent line at that point.

Formal Definition

A function f is differentiable at $x = a$ if the limit

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

exists and is finite.

Differentiability on Intervals

- If f is differentiable at each point of the open interval (a, b), then we say it is **differentiable on** (a, b)
- Similarly for open intervals of the form $(a, +\infty)$, $(-\infty, b)$, and $(-\infty, +\infty)$
- In the last case, we say that f is **differentiable everywhere**

Geometric Meaning

- Differentiable = Smooth curve with no sharp corners
- Non-differentiable = Sharp turns, cusps, vertical tangents, or discontinuities

Examples of Differentiability

Example

Determine if $f(x) = |x|$ is differentiable at $x = 0$.

Solution

$$\text{Left-hand derivative: } \lim_{h \to 0^-} \frac{|0+h| - |0|}{h} = \lim_{h \to 0^-} \frac{-h}{h} = -1$$

$$\text{Right-hand derivative: } \lim_{h \to 0^+} \frac{|0+h| - |0|}{h} = \lim_{h \to 0^+} \frac{h}{h} = 1$$

Since left-hand derivative $\neq$ right-hand derivative, $f(x) = |x|$ is **not differentiable** at $x = 0$.

Example

Determine if $f(x) = x^2$ is differentiable at $x = 1$.

Solution

$$f'(1) = \lim_{h \to 0} \frac{(1+h)^2 - 1^2}{h} = \lim_{h \to 0} \frac{1 + 2h + h^2 - 1}{h} = \lim_{h \to 0} (2 + h) = 2$$

Limit exists and is finite, so $f(x) = x^2$ is **differentiable** at $x = 1$.

Relationship: Differentiability and Continuity

Important Theorem

If a function is differentiable at $x = a$, then it must be continuous at $x = a$.

Proof

If f is differentiable at a, then $\lim_{h\to 0} \frac{f(a+h)-f(a)}{h} = f'(a)$ exists.

$$\lim_{h\to 0}[f(a+h) - f(a)] = \lim_{h\to 0}\left[\frac{f(a+h) - f(a)}{h} \cdot h\right] = f'(a) \cdot 0 = 0$$

So $\lim_{h\to 0} f(a+h) = f(a)$, which means f is continuous at a.

Warning!

The converse is **not true**: A function can be continuous but not differentiable.

- $f(x) = |x|$ is continuous at $x = 0$ but not differentiable there
- $f(x) = \sqrt[3]{x}$ is continuous at $x = 0$ but has a vertical tangent

Basic Differentiation Rules

Power Rule

$$\frac{d}{dx}[x^n] = nx^{n-1} \quad \text{for any real number } n$$

Constant Rule

$$\frac{d}{dx}[c] = 0 \quad \text{where } c \text{ is a constant}$$

Constant Multiple Rule

$$\frac{d}{dx}[cf(x)] = c \cdot f'(x) \quad \text{where } c \text{ is a constant}$$

Sum/Difference Rule

$$\frac{d}{dx}[f(x) \pm g(x)] = f'(x) \pm g'(x)$$

Worked Examples: Basic Rules

Example

Find the derivative of $f(x) = 3x^4 - 2x^2 + 5x - 7$

Solution

$$\begin{aligned} f'(x) &= \frac{d}{dx}[3x^4] - \frac{d}{dx}[2x^2] + \frac{d}{dx}[5x] - \frac{d}{dx}[7] \\ &= 3 \cdot 4x^{4-1} - 2 \cdot 2x^{2-1} + 5 \cdot 1x^{1-1} - 0 \\ &= 12x^3 - 4x + 5 \end{aligned}$$

Example

Find the derivative of $g(x) = 4\sqrt{x} - \frac{3}{x^2}$

Solution

$$g(x) = 4x^{1/2} - 3x^{-2}$$

$$g'(x) = 4 \cdot \frac{1}{2}x^{-1/2} - 3 \cdot (-2)x^{-3}$$

$$= 2x^{-1/2} + 6x^{-3} = \frac{2}{\sqrt{x}} + \frac{6}{x^3}$$

Application: Horizontal Tangent Lines

Find all points where $f(x) = x^3 - 3x^2 - 9x + 5$ has a horizontal tangent line.

1. Find derivative: $f'(x) = 3x^2 - 6x - 9$
2. Set derivative equal to 0 (horizontal tangent means slope $= 0$):

$$3x^2 - 6x - 9 = 0$$

3. Solve: $x^2 - 2x - 3 = 0 \Rightarrow (x-3)(x+1) = 0 \Rightarrow x = 3, -1$
4. Find y-coordinates:

$$f(3) = 27 - 27 - 27 + 5 = -22$$
$$f(-1) = -1 - 3 + 9 + 5 = 10$$

5. Points with horizontal tangents: $(3, -22)$ and $(-1, 10)$

Application: Area of Triangle Formed by Tangent Line

Find the area of the triangle formed by the coordinate axes and the tangent line to $y = \frac{1}{x}$ at $x = 2$.

1. Point of tangency: When $x = 2$, $y = \frac{1}{2}$, so point is $(2, \frac{1}{2})$
2. Find derivative: $y' = -\frac{1}{x^2}$, so slope at $x = 2$ is $m = -\frac{1}{4}$
3. Equation of tangent line:

$$y - \frac{1}{2} = -\frac{1}{4}(x - 2) \Rightarrow y = -\frac{1}{4}x + 1$$

4. Find intercepts:
   - x-intercept: Set $y = 0 \Rightarrow 0 = -\frac{1}{4}x + 1 \Rightarrow x = 4$
   - y-intercept: $x = 0 \Rightarrow y = 1$
5. Area of triangle: $\frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 4 \times 1 = 2$

Higher Derivatives

Definition

The **second derivative** is the derivative of the first derivative:

$$f''(x) = \frac{d}{dx}[f'(x)]$$

The **third derivative** is the derivative of the second derivative, and so on.

Notation

- First derivative: $f'(x)$, y', or $\frac{dy}{dx}$
- Second derivative: $f''(x)$, y'', or $\frac{d^2y}{dx^2}$
- Third derivative: $f'''(x)$, y''', or $\frac{d^3y}{dx^3}$
- nth derivative: $f^{(n)}(x)$ or $\frac{d^ny}{dx^n}$

Example

Find the first three derivatives of $f(x) = 4x^3 - 2x^2 + x - 5$

Solution

$$f'(x) = 12x^2 - 4x + 1$$
$$f''(x) = 24x - 4$$
$$f'''(x) = 24$$

Product Rule

If $y = f(x)g(x)$, then

$$\frac{dy}{dx} = f'(x)g(x) + f(x)g'(x)$$

Mnemonic

"First times derivative of second, plus second times derivative of first"

Find the derivative of $y = (x^2 + 1)(3x - 2)$

$$f(x) = x^2 + 1, \qquad f'(x) = 2x$$
$$g(x) = 3x - 2, \qquad g'(x) = 3$$
$$\begin{aligned} y' &= f'(x)g(x) + f(x)g'(x) \\ &= (2x)(3x - 2) + (x^2 + 1)(3) \\ &= 6x^2 - 4x + 3x^2 + 3 = 9x^2 - 4x + 3 \end{aligned}$$

Quotient Rule

If $y = \frac{f(x)}{g(x)}$, then

$$\frac{dy}{dx} = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$$

Mnemonic

"Low d-high minus high d-low, over low squared"

Find the derivative of $y = \frac{x^2+1}{2x-3}$

$$f(x) = x^2 + 1, \qquad f'(x) = 2x$$
$$g(x) = 2x - 3, \qquad g'(x) = 2$$
$$y' = \frac{(2x)(2x-3) - (x^2+1)(2)}{(2x-3)^2}$$
$$= \frac{4x^2 - 6x - 2x^2 - 2}{(2x-3)^2} = \frac{2x^2 - 6x - 2}{(2x-3)^2}$$

Derivative of Reciprocal

Special Case of Quotient Rule

If $y = \frac{1}{g(x)}$, then

$$\frac{dy}{dx} = -\frac{g'(x)}{[g(x)]^2}$$

Example

Find the derivative of $y = \frac{1}{x^2+1}$

Solution

$$g(x) = x^2 + 1, \qquad g'(x) = 2x$$

$$y' = -\frac{2x}{(x^2+1)^2}$$

Example

Find the derivative of $y = \frac{1}{\sqrt{x}}$

Solution

$$y = x^{-1/2}$$

$$y' = -\frac{1}{2}x^{-3/2} = -\frac{1}{2x^{3/2}} = -\frac{1}{2x\sqrt{x}}$$

Practice Problems: Basic Level

Differentiability and Basic Rules

1. Determine if $f(x) = \sqrt{x}$ is differentiable at $x = 0$
2. Find the derivative of $f(x) = 5x^3 - 2x^2 + 7x - 4$
3. Find where $f(x) = x^2 - 4x + 3$ has horizontal tangent lines
4. Find the second derivative of $g(x) = 3x^4 - 2x^2 + x$

Product and Quotient Rules

1. Find the derivative of $y = (2x + 1)(x^2 - 3)$
2. Find the derivative of $y = \frac{x}{x^2+1}$

Practice Problems: Intermediate Level

Applications and Combined Rules

1. Find the area of the triangle formed by the coordinate axes and the tangent line to $y = x^2$ at $x = 1$
2. Find the derivative of $y = \frac{(x+1)(x-2)}{x^2+1}$
3. A ball is thrown upward with height $h(t) = -16t^2 + 64t + 5$. Find:
   - When the ball reaches its maximum height
   - The velocity when it hits the ground
4. Find all points where $f(x) = x^3 - 6x^2 + 9x + 1$ has horizontal tangents

Practice Problems: Advanced Level

Challenging Problems

1. Find the derivative of $y = \frac{x^2\sqrt{x+1}}{(2x-1)^2}$
2. A rectangle is inscribed under the curve $y = 4 - x^2$ with its base on the x-axis. Find the maximum possible area.
3. Prove that if $f(x) = x^n$ for a positive integer n, then $f^{(n)}(x) = n!$ and $f^{(n+1)}(x) = 0$

Key Formulas Summary

Basic Differentiation Rules

- Power Rule: $\frac{d}{dx}[x^n] = nx^{n-1}$
- Constant Rule: $\frac{d}{dx}[c] = 0$
- Constant Multiple: $\frac{d}{dx}[cf(x)] = cf'(x)$
- Sum/Difference: $\frac{d}{dx}[f(x) \pm g(x)] = f'(x) \pm g'(x)$

Product and Quotient Rules

- Product: $\frac{d}{dx}[f(x)g(x)] = f'(x)g(x) + f(x)g'(x)$
- Quotient: $\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$
- Reciprocal: $\frac{d}{dx}\left[\frac{1}{g(x)}\right] = -\frac{g'(x)}{[g(x)]^2}$

Important Theorems

- Differentiable $\Rightarrow$ Continuous
- Horizontal tangent when $f'(x) = 0$
- Higher derivatives: $f''(x)$, $f'''(x)$, etc.

Basic Trigonometric Derivatives

Fundamental Derivatives

$$\frac{d}{dx}[\sin x] = \cos x$$
$$\frac{d}{dx}[\cos x] = -\sin x$$
$$\frac{d}{dx}[\tan x] = \sec^2 x$$
$$\frac{d}{dx}[\cot x] = -\csc^2 x$$
$$\frac{d}{dx}[\sec x] = \sec x \tan x$$
$$\frac{d}{dx}[\csc x] = -\csc x \cot x$$

Memory Aid

- Sine and cosine derivatives cycle: $\sin \to \cos \to -\sin \to -\cos \to \sin$
- Derivatives of co-functions have negative signs
- Secant and cosecant follow patterns with their respective co-functions

Worked Examples: Basic Derivatives

Example

Find the derivative of $f(x) = 3\sin x - 2\cos x$

Solution

$$\begin{aligned} f'(x) &= 3\frac{d}{dx}[\sin x] - 2\frac{d}{dx}[\cos x] \\ &= 3(\cos x) - 2(-\sin x) \\ &= 3\cos x + 2\sin x \end{aligned}$$

Example

Find the derivative of $g(x) = x^2 \tan x$

Solution

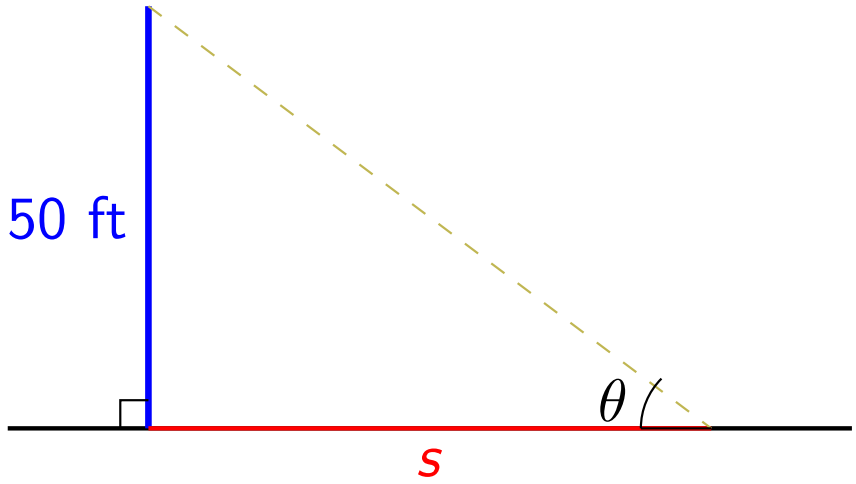
Using the product rule:

$$\begin{aligned} g'(x) &= \frac{d}{dx}[x^2] \cdot \tan x + x^2 \cdot \frac{d}{dx}[\tan x] \\ &= (2x)(\tan x) + x^2(\sec^2 x) \\ &= 2x \tan x + x^2 \sec^2 x \end{aligned}$$

Application: Flagpole Shadow Problem

Example

On a sunny day, a 50 ft flagpole casts a shadow that changes with the angle of elevation of the Sun. Let s be the length of the shadow and θ the angle of elevation of the Sun. Find the rate at which the length of the shadow is changing with respect to θ when $\theta = 45°$. Express your answer in units of feet/degree.



Solution

1. From the right triangle: $\tan\theta = \frac{50}{s} \Rightarrow s = \frac{50}{\tan\theta} = 50\cot\theta$
2. Differentiate with respect to θ:

$$\frac{ds}{d\theta} = 50 \cdot (-\csc^2\theta) = -50\csc^2\theta$$

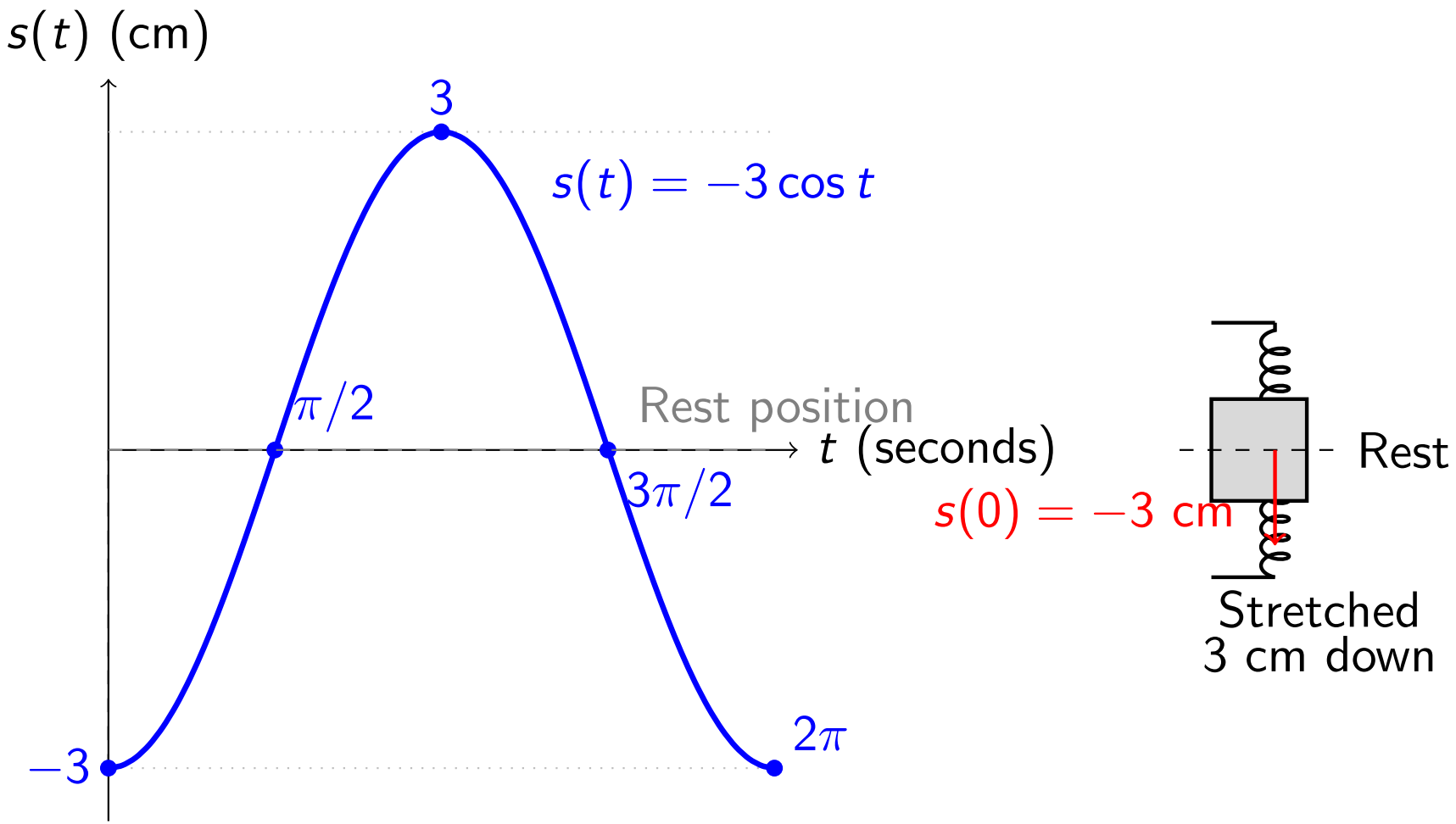
3. Convert $\theta = 45^\circ$ to radians: $\theta = \frac{\pi}{4}$ radians
4. Evaluate: $\frac{ds}{d\theta} = -50\csc^2\left(\frac{\pi}{4}\right) = -50 \cdot (\sqrt{2})^2 = -100$ ft/radian
5. Convert to ft/degree: $-100 \cdot \frac{\pi}{180} \approx -1.745$ ft/degree

The shadow length decreases at about 1.745 ft per degree when $\theta = 45^\circ$.

Application: Spring Motion Problem

Example

A spring with an attached mass is stretched 3 cm beyond its rest position and released at time $t = 0$. The position function is $s = -3\cos t$ where s is in centimeters and t is in seconds. Find the velocity function and discuss the motion of the attached mass.


$s(t)$ (cm)
3
$s(t) = -3\cos t$
$\pi/2$
Rest position
t (seconds)
$3\pi/2$
2π
-3
$s(0) = -3$ cm
Rest
Stretched
3 cm down

Solution

1. Velocity is the derivative of position:

$$v(t) = \frac{ds}{dt} = \frac{d}{dt}[-3\cos t] = -3(-\sin t) = 3\sin t$$

2. Initial conditions:
   - At $t = 0$: $s(0) = -3\cos 0 = -3$ cm (3 cm below rest)
   - $v(0) = 3\sin 0 = 0$ cm/s (starting from rest)
3. Motion analysis:
   - The mass oscillates between -3 cm and $+3$ cm
   - Maximum speed: $|v(t)|_{max} = 3$ cm/s when $\sin t = \pm 1$
   - Zero speed at turning points: $s = \pm 3$ cm
   - Period: 2π seconds

Higher Order Trigonometric Derivatives

Example

Find the first four derivatives of $f(x) = \sin x$

$$
\begin{aligned}
f(x) &= \sin x \\
f'(x) &= \cos x \\
f''(x) &= -\sin x \\
f'''(x) &= -\cos x \\
f^{(4)}(x) &= \sin x
\end{aligned}
$$

Pattern Observation

The derivatives of $\sin x$ follow a four-step cycle:

$$\sin x \to \cos x \to -\sin x \to -\cos x \to \sin x \to \cdots$$

Practice Problems: Basic Level

Basic Derivatives

1. Find the derivative of $f(x) = 2\sin x - 3\cos x + \tan x$
2. Find the derivative of $g(x) = x\cos x$
3. A particle moves according to $s(t) = 4\sin t$. Find its velocity and acceleration functions.

Practice Problems: Intermediate Level

Applications and Combined Rules

1. A ladder 10 ft long rests against a vertical wall. If the bottom of the ladder slides away from the wall at 1 ft/s, how fast is the angle between the ladder and the ground changing when the bottom is 6 ft from the wall?
2. Find the equation of the tangent line to $y = x \sin x$ at $x = \pi$
3. Find all points on the curve $y = \sin x + \cos x$ where the tangent line is horizontal
4. The position of a pendulum is given by $\theta(t) = 0.5 \cos(2t)$. Find its angular velocity and acceleration.

Practice Problems: Advanced Level

Challenging Problems

1. A water wave has height $h(x, t) = 2\sin(3x - 4t)$ meters. Find the rate of change of height with respect to time at a fixed position $x = 2$ m.
2. Find the derivative of $y = \dfrac{\sin x \cos x}{1 + \tan x}$
3. A Ferris wheel with radius 20 m rotates at 1 radian per second. If a seat starts at the bottom, find the rate at which the seat is rising when it is 10 m above the ground.
4. Show that $y = A\sin(\omega t) + B\cos(\omega t)$ satisfies the differential equation $\frac{d^2y}{dt^2} + \omega^2 y = 0$

More Application Problems

Physics and Engineering Applications

1. An alternating current is given by $I(t) = 5\sin(120\pi t)$ amperes. Find the rate of change of current at $t = 0.01$ seconds.
2. The voltage in a circuit is $V(t) = 220\cos(100\pi t)$ volts. Find the instantaneous rate of change of voltage.
3. A sound wave has pressure $P(x, t) = 0.02\sin(300x - 1000t)$ Pascals. Find how fast the pressure is changing at a point 2 m from the source after 0.1 seconds.
4. A piston moves according to $x(t) = 0.1\cos(20t)$ meters. Find its maximum speed and acceleration.

Important Applications

- Physics: Simple harmonic motion, waves, pendulums
- Engineering: Alternating current, mechanical vibrations
- Geometry: Related rates problems with angles
- Real-world: Shadow lengths, ladder problems, Ferris wheels

Problem-Solving Tips

- Convert between degrees and radians carefully in applied problems
- Look for patterns in higher-order derivatives